LITHIUM-ION POWER
OEM SOLUTIONS
FUTURE OF ENERGY STORAGE

Goal Diggers

Rachit Makkar

Tarun Kumar

Prateek Sangwan

Mouly Mundra

MARKSOC
DTU

1 Growing Power Backup Demand in India

Frequent power outages in Tier 2 & Tier 3 cities
Rising electricity instability due to urban expansion
Small businesses relying on uninterrupted power

Frequent outages, rising urbanization, and increasing digital dependence are driving strong demand for reliable backup solutions. Small businesses and households now view uninterrupted power as essential for daily productivity and stability.

2 Shift from Lead-Acid to Lithium Technology

Longer battery life (8–12 years vs 3–5 years)
Faster charging
Lower long-term cost
Environmental concerns around acid & disposal

Consumers prefer maintenance-free lithium batteries due to longer lifespan, faster charging, lower lifetime cost, and environmental benefits, signaling a clear market transition from traditional lead-acid technology.

3 Internal Capability Advantage

Strong R&D and manufacturing base (B2B expertise)
In-house lithium technology
Existing industrial credibility

Strong R&D, in-house lithium expertise, and established manufacturing credibility reduce technological risk and provide competitive advantage over new market entrants.

4 Policy & Industry Tailwinds

Push toward renewable & solar integration
Growing home solar adoption
Government support for energy-efficient solutions
Lithium aligns with future-ready energy ecosystem.

Rising solar adoption and government support for energy-efficient solutions favor lithium-based systems, aligning with India's transition toward a sustainable, renewable energy ecosystem.

5 Gap in "Lithium-Only Smart Inverter" Branding

Established brand strongly positioned around lead-acid
Lithium models exist but:
Poor digital branding
Weak differentiation

Limited strong branding in lithium-only inverters creates opportunity to position a differentiated, modern smart-home-focused brand with clear messaging and digital-first marketing strategy.

Graphical Segmentation

Primary Focus:
Tier 1 Cities (Delhi NCR, Mumbai, Bangalore, Pune)
Tier 2 Cities (Lucknow, Jaipur, Indore, Patna, etc.)
Why?
Frequent outages in Tier 2
Higher paying capacity in Tier 1
Dense housing → easier service coverage

Behavioral Segmentation

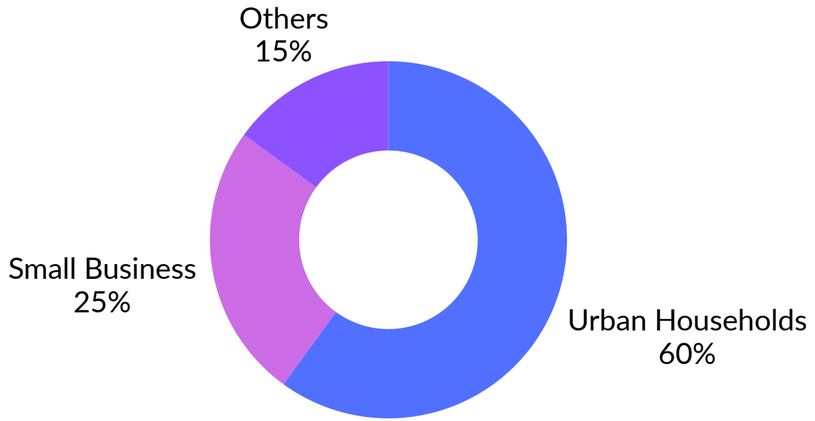
- Have faced frequent power cuts
- Recently replaced lead-acid batteries
- Hate maintenance (water topping, leakage)
- Value long-term savings over upfront cost
- Work from home or run home offices

Demographic Segmentation

- Tech-aware
- Environment-conscious
- Prefers modern, premium products
- Values hassle-free ownership
- Thinks long-term, not cheapest option

Demographic Segmentation

Segment	Income Level	Home Type	Buying Capacity
Upper Middle Class	₹8–15 LPA	Owned Flats / Houses	High
Middle Class	₹5–8 LPA	Owned Homes	Moderate
Small Commercial Users	₹6–12 LPA	Shops / Clinics	High Need



INSTEAD OF TARGETING THE ENTIRE INVERTER MARKET, WE FOCUS ON HIGH-VALUE, RELIABILITY-DRIVEN URBAN CUSTOMERS WHO ALIGN WITH OUR PREMIUM LITHIUM POSITIONING AND MARGIN GOALS.

Primary Target Persona

"Urban Reliability Seeker"

- 30–45 years old
- Family household
- Work-from-home professional
- Wants uninterrupted power for AC, WiFi, laptop
- Willing to pay more for peace of mind

Secondary Target Persona

"Small Business Continuity Owner"

- Runs clinic / salon / shop
- Power cut = revenue loss
- Values fast charging & reliability
- Ready to invest for long-term stability

Strategic Decision

- Focus on Urban Middle & Upper Middle-Class Households first
- Avoid mass price-sensitive rural segment initially
- Avoid competing in low-cost segment

POSITIONING-

"The compact inverter designed for apartments facing frequent power cuts."

- **Given its faster recharge, compact design, and consistent backup performance, Powergate should be positioned as a practical upgrade over traditional lead-acid systems for NCR apartments facing frequent power interruptions.**

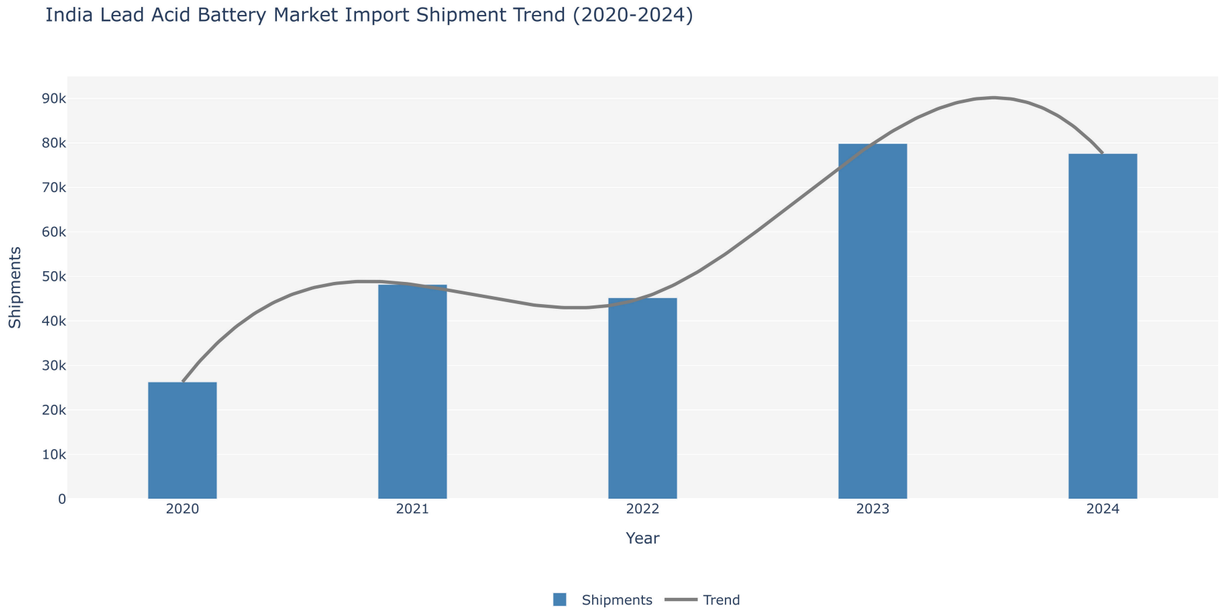
- **Rather than competing on price, it should be positioned as a modern replacement for lead-acid systems, optimized for NCR's apartment living conditions and intermittent power supply.**

VALUE PROPOSITION-

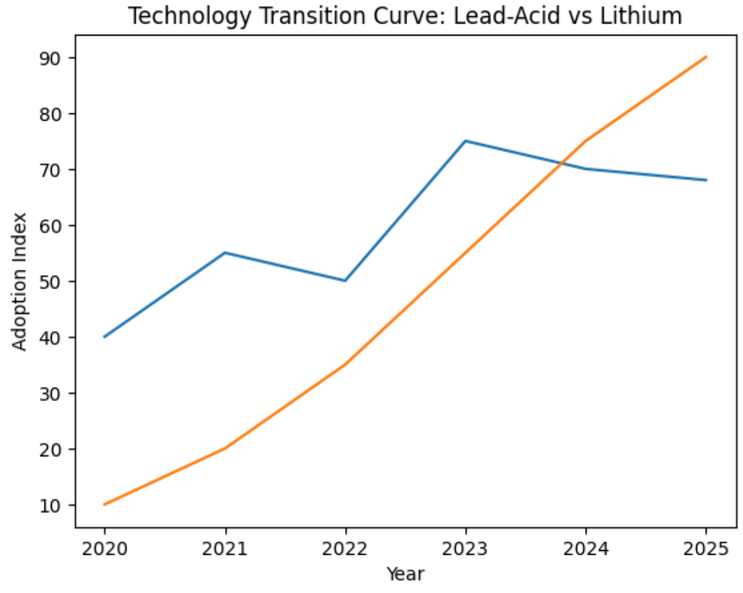
why should a lead-acid battery inverter user switch to powergate

- **faster recharge, ready for multiple outages, lead-acid batteries generally use 10-12 hours to recharge, lithium ion battery takes 4-5hours, so at places with multiple outages it provides better output.**
- **compact design-Urban households in Noida and the NCR region face significant space constraints due to high-density apartment living making compact and wall-mountable energy storage solutions far more practical than bulky lead-acid battery setups.**
- **local service- reduces trust barrier. consumer fear services and warranty issues. Local factory backed service provides reliability**

- **The India lead acid battery market import shipment demonstrates that the growth rate between 2023 and 2024 experienced a slight decline, indicating a potential stabilization or slowdown in the market momentum towards the end of the period.**



- **the consistant growth trend from 2020-2023 shows the increasing demands of inverter batteries, however it reaches it's saturation point in the year 2024, the next step have to be the LITHIUM ION batteries. despite the constant demand of lead-acid battery, it faces issues related to maintenance, space consumption and shorter lifespan- creating a gap which can easily be filled by lithium ion batteries**



Distribution through OEM Solutions website

selling via own website :

- full margins
- user data insights
- Reducing fixed cost

Using factory space as INVENTORY utilizes the given assets with “Zero incremental warehousing cost”

location: F-35, Sector-8, Noida (G.B. Nagar), UP – 201301

Experience Centers" (existing high-end electronics boutiques near noida)

No Inventory Risk: These partners display one "Compact & Elegant" unit so customers can see the product. The partners don't hold stock; they earn a small "referral fee" for every sale made through a QR code on the display unit

Category	D2C Model (Website)	Offline Dealership Model
COGS (Production)	75%	75%
Logistics/Marketing	5%	5%
Middleman Margin	0%	15% (Dealer Cut)
Net Profit	20% (Target Hit)	5% (Target Missed)

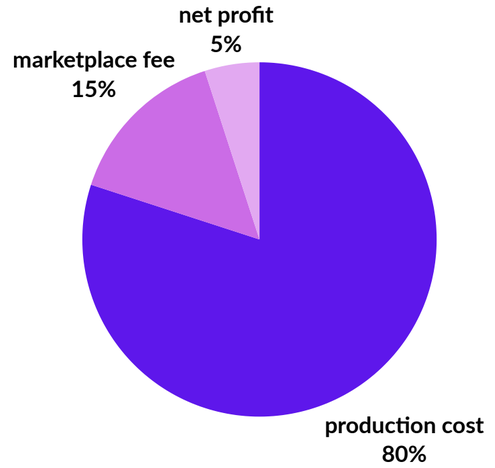
COMPARSION B/W OFLINE AND ONLINE SALES MODEL

NO NEED OF OFFLINE DEALERSHIP

- Inventory Risk: Traditional dealers often expect "Credit Periods"(you give them the stock, they pay you 30-60 days later).
- For a bootstrapped company, this kills your cash flow.
- In the inverter market, offline dealers are usually "brand loyal" to giants like Luminous or Microtek
- because of long-standing relationships and high-volume rebates.
- Only have 1,000 units, spreading them across 50-100 dealers means each dealer only has 10 units.
- Stock Outs: If one dealer sells out and another has zero sales, your inventory is "stuck" in the wrong place.
- Digital Control: Selling via your own website (D2C) allows you to manage all 1,000 units from a single central warehouse in Noida, ensuring every unit goes to a confirmed, paying customer.

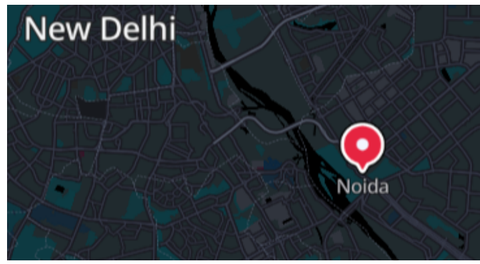
NO MARKETPLACES (AMAZON & FLIPKART)

- No marketplace/brokerage fees
- Increased margins
- No refund/return exploitations
- No forced competitive pricing
- Pricing control
- Customer data ownership
- Better demand forecasting
- Lower working capital blockage
- Review protection
- Brand control

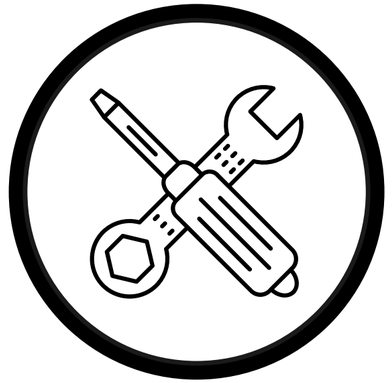


This data does’nt includes the compensations of returns and refunds ,
if included the 5% net profit will also be consumed

Main target costomers-
residents near Noida & Delhi-Ncr



- **Localized Factory Support**: For the initial 1,000 units, use the existing Noida-based technical team to handle installations and repairs in the Delhi-NCR region, ensuring high reliability during the "demand validation" phase.
- **Third-Party Service Partnerships:** To avoid "heavy fixed investments," the company should partner with "Service-as-a-Service" providers (like Onsitego or Jeeves) for regions outside of Noida to maintain an asset-light footprint
- **Industry-Leading Coverage:** Offer a 5-year comprehensive warranty on both the 1100 VA inverter and the 100Ah battery. Most competitors only offer 2–3 years on the inverter..
- **Hyper-Local Noida Support:** Use the existing R&D and manufacturing team in Noida to handle the first 1,000 installations, ensuring a "Factory-to-Floor" expertise.
- **Direct Video Support:** Implement a first-level troubleshooting protocol via video calls
- directly from the factory's service experts to resolve minor issues without a physical visit.
- **White-Glove Delivery:** Instead of a courier dropping a 40kg box at the door, the delivery is bundled with a certified technician who performs the installation and a "health check" of the home's wiring.
- **Modular Product Design:** Design the "Powergate" system so that the 100Ah LiFePO4 battery or the 1100 VA inverter components can be easily replaced or swapped if a major failure occurs, reducing repair time.



- **The "Unboxing" Education: The technician spends 10 minutes showing the user the "Maintenance-Free" aspect—pointing out that there is no acid to check or water to top up.**
- **Remote Diagnostics: Use the "Smart" capability of the inverter to provide over-the-phone troubleshooting, resolving 40% of issues without a physical visit (saving on OPEX).**
- **Battery Buy-Back: Create a "Green Exchange" program where the company collects the old lead-acid battery for recycling, lowering the customer's entry cost while building brand loyalty.**

Contribution Margin Analysis

Contribution Per Unit:
Selling Price – Variable Cost
₹24,000 – ₹18,700 = ₹5,300
Contribution Margin %:
5,300 / 24,000 = 22%

Break-Even Analysis

Assume Fixed Costs (example):
Branding & Website Setup = ₹10,00,000
Initial Marketing Push = ₹5,00,000
Admin & Overheads = ₹5,00,000
Total Fixed Cost = ₹20,00,000
Break-even Units = Fixed Cost / Contribution per Unit
= 20,00,000 / 5,300 ≈ 377 units

Sensitivity Analysis

What If Price Drops to ₹23,000?
Contribution = 23,000 – 18,700 = 4,300
Margin = 18.7%
Still viable but tight.
What If CAC increases by ₹500?
New total cost = ₹19,200
Contribution = ₹4,800
Margin ≈ 20%
Still within target.

UNIT ECONOMICS

	COMPONENTS	COST
1	MANUFACTURING	16,500
2	CAC	1300
3	DELIVERY AND INSTALL	900
4	TOTAL COST	18700

COST BREAKDOWN-

- **SELLING PRICE-24,000**
- **TOTAL COST PER UNIT -18,700**
- **CONTRIBUTION PER UNIT-5,300**
- **MARGIN=22%**

- **REVENUE=2,40,00,000**
- **TOTAL COST=1,80,00,000**
- **GROSS PROFIT=60,00,000**

- **Production costs COGS-** including battery cells, BMS, casing, assembly, packaging
- **Customer Acquisition Cost-** advertisments, society campaigns and refferal incentives
- **Delivery and installations**- local delivery fees, electrician commisions

Working Capital Efficiency

D2C Model Benefits:
No marketplace commission
Faster cash realization
Factory-to-customer dispatch
Low inventory holding
Working Capital Cycle = Short
Cash flow risk = Controlled

Marketing Efficiency

CAC = ₹1,300
Add projection:
If digital optimization reduces CAC to ₹1,000:
Contribution increases to ₹5,600
Margin improves to 23.3%
Shows scalability potential.

Phase 1: Awareness & Lead Capture (Month 1–2)

Objective:
Build demand ONLY in Noida & South Delhi.

Actions:

1. Hyper-Local Digital Ads
Instagram & Facebook Ads
10 km radius targeting around:
Noida Sector 8 (factory area)
Experience Centers
Focus messaging:

2. Society-Level Activations
RWAs (Resident Welfare Associations)
Apartment notice boards
Demo days in high-rise societies
On-ground booth with display unit

3. Pre-Order Consultation Model
Before purchase:
2-minute phone verification
Load requirement confirmation
Installation scheduling
KPI:
1,500–2,000 verified leads
300 strong purchase-intent customers

Phase 2: Conversion Push (Month 3–4)

Objective:
Sell first 300–500 units

1. Experience Center Strategy
As mentioned in your Distribution Slide:
Display unit in high-end electronics boutiques
QR code-based referral model
₹1,000 in-store exclusive discount
Dealer earns 5% referral fee (no inventory risk)
This becomes a marketing + distribution hybrid model.

2. Influencer Strategy (Local, Not National)
NCR-based tech YouTubers
Apartment-living vloggers
“Home upgrade” Instagram creators
Focus:
Compact size
Fast charging
Maintenance-free

3. Testimonial Campaign
After first 100 installations:
Record real apartment installation videos
Show side-by-side comparison vs lead-acid
Social proof builds trust.

Optimization & Scale to 1000 Units (Month 5–6)

Objective:
Sell remaining inventory while reducing CAC

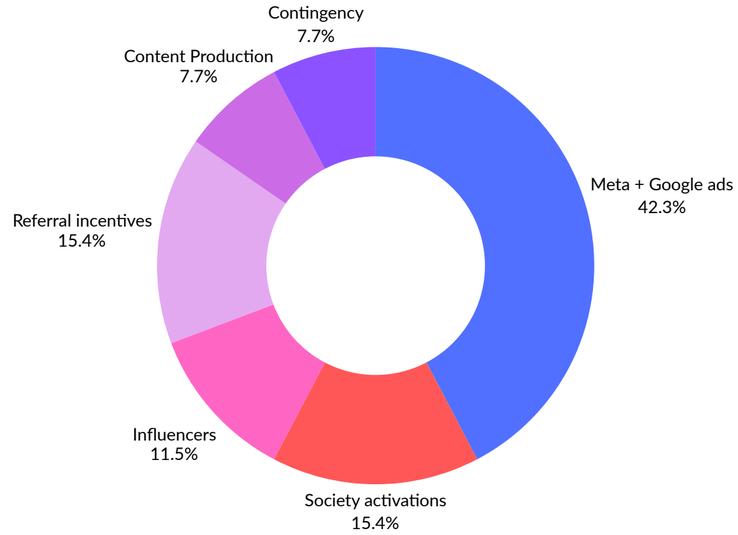
1. Retargeting Campaigns
Retarget website visitors
Show EMI option ads
Highlight 5-year warranty

2. Referral Program
Offer:
₹1,000 referral benefit per successful installation
Apartment clusters amplify word-of-mouth.
3. Green Exchange Campaign

Promote:
“Exchange your old lead-acid battery & upgrade to lithium.”
Aligns with:
Environmental positioning
Solar-ready messaging

Marketing Budget Allocation (Aligned to CAC ₹1,300)

Total Marketing Budget = ₹13,00,000
Breakdown:
₹5.5L – Hyper-local Meta + Google Ads
₹2L – Society Activations
₹1.5L – Influencers (NCR-focused)
₹2L – Referral incentives
₹1L – Content production
₹1L – Contingency
Controlled, practical, bootstrapped.



RISK ASSESMENT

Risk 1:

You pay the dealer a 5% referral fee, but a Luminous/other distributor offers them 15% to "switch" the customer at the last second.
The "Pro" Mitigation: QR-Locked Exclusive Pricing. * Practical Solution: The display unit in the shop has a QR code that gives a ₹1,000 "In-Store Only" discount. The customer cannot get this price on the open web. This forces the customer to scan the dealer's specific code, locking in the dealer's 5% commission and ensuring they don't try to sell a competitor's product.

Risk 2:

High Customer Acquisition Cost (CAC) on D2C. Without Amazon's traffic, finding 1,000 buyers might be expensive.
Practical Solution: Hyper-Local Geofencing. Instead of national ads, run highly targeted Instagram/Facebook ads only within a 10km radius of your "Experience Centers" in Noida and South Delhi. This ensures every ad rupee spent is likely to lead to a physical product demo.

Risk 3:

Low Motivation for Referral Partners. Why would a shop owner refer you if they don't get a huge dealer margin?
Practical Solution: Zero Inventory Burden. Contrast your model for them: They get a 5% "Pure Profit" referral fee without the risk of buying stock, blocking shop space, or managing after-sales. It's "Passive Income" for the shop owner, making it an easy sell for you.

Risk 4:_

High Return/Refund Rates on Marketplaces. Marketplaces have "no-questions-asked" return policies that could bleed a startup dry on shipping a 40kg unit.
Practical Solution: The "Self-Vetted" Lead. By selling only through your website, you can implement a "Pre-order Consultation" step. A quick 2-minute phone verification ensures the customer's home load matches the 1100 VA inverter capacity before shipping, reducing "fitment" returns to near zero.

Hyper-Local Noida Support: Use the existing R&D and manufacturing team in Noida to handle the first 1,000 installations, ensuring a "Factory-to-Floor" expertise.

THANK YOU